Next Step Assignment: Pruning, Cross-Validation, and XGBoost

Part A: Questions after basic Decision Tree training

Before improving the model, answer these questions from your first Decision Tree results:

1. What was the train score and test score for Decision Tree Regression?
2. Was the training error much lower than the testing error?
3. Did the tree become too deep?
4. How many leaf nodes were created?
5. Did you observe small leaves with very few samples?
6. In the regression tree, which feature appeared near the root node?
7. In the classification tree, which features appeared important?
8. Did the unpruned tree look easy to explain to a business user?
9. Was the model overfitting, underfitting, or generalizing well?
10. What metric did you trust more: train score or test score? Why?

Part B: Think before improving

Now answer:

1. If the tree is too deep, which hyperparameter can control it?
2. If the tree has very small leaf nodes, which hyperparameter can control it?
3. If we want only a limited number of business segments, which hyperparameter can control it?
4. Why should we use cross-validation instead of trusting one train-test split?
5. Why might XGBoost perform better than a single Decision Tree?

Part C: Improvement plan

Now improve both Regression and Classification models using three levels:

Level 1: Pre-pruning

Train Decision Tree models again using:

- `max_depth`
- `min_samples_split`
- `min_samples_leaf`
- `max_leaf_nodes`

Compare before pruning and after pruning.

For Regression, compare:

- MAE
- MSE
- RMSE
- R2 score
- tree depth
- number of leaves

For Classification, compare:

- accuracy
- precision
- recall
- F1-score
- confusion matrix
- tree depth
- number of leaves

Level 2: Post-pruning

Use cost-complexity pruning with `ccp_alpha`.

Steps:

1. Train a full Decision Tree.

2. Generate possible `ccp_alpha` values.

3. Train multiple trees using different `ccp_alpha` values.

4. Compare train score and test score.

5. Select the alpha that gives stable test performance with fewer leaves.

6. Explain whether pruning reduced overfitting.

Level 3: Cross-validation

Use cross-validation to avoid depending on a single train-test split.

Steps:

1. Use 5-fold cross-validation.

2. Compare average CV score for unpruned and pruned trees.

3. Use `GridSearchCV` to search the best hyperparameters.

4. Report the best parameters.

5. Report the best cross-validation score.

6. Evaluate the final tuned model on the test set.

Level 4: XGBoost

Train XGBoost models for both tasks:

- `XGBRegressor` for regression
- `XGBClassifier` for classification

Compare XGBoost against:

- unpruned Decision Tree
- pruned Decision Tree
- cross-validated Decision Tree

For Regression, compare:

- RMSE
- MAE
- R2

For Classification, compare:

- accuracy
- precision
- recall
- F1-score
- confusion matrix

Part D: Final comparison table

Create one final table for Regression:

Model	Train RMSE	Test RMSE	Test MAE	Test R2	Depth	Leaves	Remark
Unpruned Decision Tree							
Pre-pruned Decision Tree							
Post-pruned Decision Tree							
Cross-validated Decision Tree							
XGBoost Regressor					NA	NA	

Create one final table for Classification:

Model	Accuracy	Precision	Recall	F1-score	Depth	Leaves	Remark
Unpruned Decision Tree							
Pre-pruned Decision Tree							
Post-pruned Decision Tree							
Cross-validated Decision Tree							
XGBoost Classifier					NA	NA	

Part E: Final student conclusion

Write 8–10 lines answering:

1. Did pruning improve the model?
2. Did pruning reduce tree complexity?
3. Which hyperparameter had the biggest effect?
4. Did cross-validation give more reliable results?
5. Did XGBoost perform better than Decision Tree?
6. Which model would you choose for explanation?
7. Which model would you choose for prediction?
8. What business insights did you get from important features?